

Q 4: Runoff Hydrograph Analysis [15 points]

Figure below shows a storm hydrograph recorded at a stream gauging station following a significant rainfall event. The hydrograph displays the characteristic response of a small catchment to precipitation input.

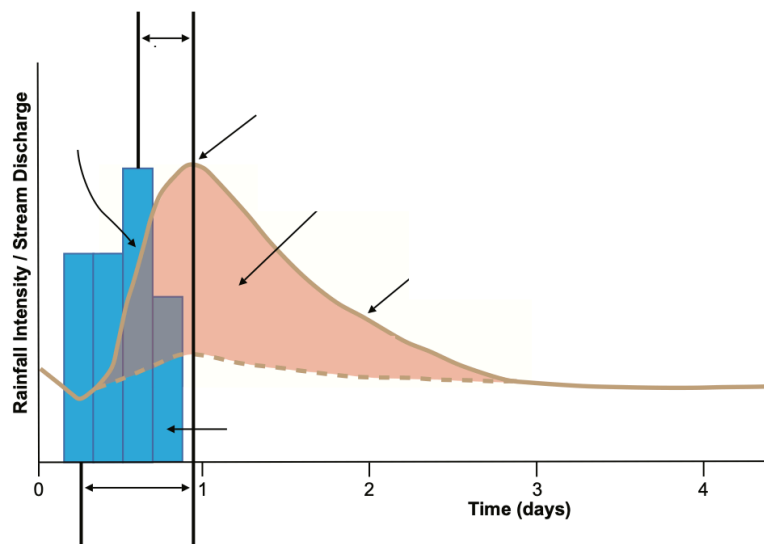


Figure 2: Storm hydrograph showing discharge response to rainfall event

Catchment characteristics:

- Catchment area: $A = 85 \text{ km}^2$
- Soil type: Clay-loam with moderate infiltration capacity
- Land use: Mixed agriculture and forest
- Antecedent conditions: Moderately wet (soil moisture at 60% of field capacity)

Tasks:

a) [2 points] On the hydrograph, clearly identify and label the following components:

- Rising limb
- Peak discharge
- Recession limb
- Baseflow (pre-storm and post-storm)

b) [3 points] Explain the hydrological processes responsible for each part of the hydrograph: